

Nigel Job Search Assistant: Build Brief

Oct 6, 2026 · @Michael Down

Purpose

Build a private web app that finds data-privacy jobs for Nigel Down every day, scores each one against his profile, and drafts a cover letter in his voice for the ones worth applying to. Nigel reviews, edits and applies himself; the system never submits an application.

Nigel is a UK data privacy professional (CIPP/E, CIPM, AIGP) who left MullenLowe Global in July 2026 after eight years as Data Privacy & Operations Director. He is currently applying by hand at a rate of three to eight roles a day: he finds a role on LinkedIn, pastes the job spec into a spreadsheet alongside his CV, LinkedIn summary and personal statement, writes a short set of prompt notes, and asks an AI assistant for a cover letter. His tracker holds 25 such rows for 29 September to 6 October 2026.

This brief replaces that manual loop with a daily pipeline and a review dashboard. It is written for an AI build agent: every requirement, constraint and decision it needs is here, and anything left open is listed at the end. The commissioning owner is Michael Down (Nigel's son), who will host and maintain the system.

Goals and non-goals

The system succeeds if Nigel opens the dashboard each morning, sees the day's relevant roles ranked by fit, and can apply to the good ones with a letter that needs only light editing.

Goals, in priority order:

1. Surface every relevant new role from LinkedIn plus two further boards within 24 hours of posting, with duplicates collapsed.
2. Score each role for fit against Nigel's profile and explain the score in two or three lines, including any hard blockers (must be a qualified solicitor, must be on site in Leeds, and so on).
3. Draft a cover letter for every role above the fit threshold, following the style guide in this brief, ready to copy or download as .docx.
4. Track what Nigel has done with each role (new, shortlisted, applied, rejected, interview) so nothing is lost and nothing is surfaced twice.
5. Be secure: only Nigel and Michael can log in, and Nigel's CV and personal data never leave the system except to the Claude API for drafting.

Non-goals for version 1:

- No automatic submission of applications, including LinkedIn Easy Apply and Indeed quick apply. This keeps Nigel in control and avoids breaching board terms of service.
- No CV rewriting. The letter adapts to the role; the CV stays as Nigel maintains it. (A per-role "CV points to emphasise" note is a cheap phase 2 addition.)
- No logging in to LinkedIn or Indeed with Nigel's accounts, and no scraping those sites directly from the app. Job data comes from third-party providers with their own API terms.
- No multi-user product. Two allow-listed users, one candidate profile.

Guiding principles:

- Human in the loop at every step that touches an employer.
- Prefer boring, hosted services over bespoke infrastructure; Michael maintains this alongside a full-time job.
- Nigel is the user, not a developer: the dashboard needs no training and works on a phone.
- Everything the AI writes is a draft and is labelled as one until Nigel marks it reviewed.

The candidate

Nigel is a senior privacy operations generalist rather than a lawyer: ten-plus years running global privacy programmes inside advertising groups, strong on DPIAs, RoPAs, vendor due diligence, transfers (SCCs/IDTAs), DSARs, training, M&A support and privacy metrics, with a project-management and data-driven-marketing background underneath.

Attribute	Value
Name	Nigel Down, nigel@nigeldown.com
Location	UK; the roles he applies for are London hybrid or UK remote (home postcode to be confirmed for the radius setting)
Most recent role	Data Privacy & Operations Director, MullenLowe Global (IPG), Jan 2018 to Jul 2026
Earlier roles	Project Director, McCann Enterprise; Senior Programme Director, Rapp; Programme Director, Proximity London; HPS/BMW; Dunnhumby (Tesco Clubcard)

Attribute	Value
Certifications	CIPP/E, CIPM, AIGP (certificates supplied); studying CIPT; IAPP Fellow of Information Privacy application pending
Regulations	UK and EU GDPR, PECR, CCPA, POPIA; familiar with ISO 27001, ISO 27701, NIST
Tools	OneTrust, Microsoft Purview, master data management, bespoke dashboards (built "Mantle" at MullenLowe)
Sectors	Advertising and marketing services, FMCG (Unilever), UK Government and NHS comms, financial services, automotive, retail
Not	A qualified solicitor. Roles that require a practising lawyer are a hard blocker; roles that prefer one are a flag, and the letter addresses it directly

Target roles

The search taxonomy below is derived from the 25 roles Nigel applied to between 29 September and 6 October 2026. It is the default; Nigel can edit every list from the dashboard settings page.

Tier	Titles and phrases to search	Examples from his applications
Core	Data Privacy Manager, Data Protection Manager, Senior Data Protection Specialist, Data Protection Officer, DPO, Head of Data Protection, Head of Privacy, Privacy Senior Manager, Privacy Operations Lead, Group Data Compliance Manager	PureGym, IG Group, Ripple, Howden, Colt, Mills & Reeve, IQ-EQ, Gallagher, Columbia Threadneedle, Forsyth Barnes (DPO)

Tier	Titles and phrases to search	Examples from his applications
Adjacent	Information Governance Manager, Data Governance Lead, Data Governance Assistant Director, AI Governance Lead, People Data & Governance Lead, Compliance Manager (data protection remit), Legal & Compliance Senior Manager, Privacy Programme Manager	EY, QH4, Lockton, Regent's University, St John & Elizabeth Hospital, Help for Heroes, Affirm, Vontier, Oliver James
Stretch (off by default)	Client Data Manager, Master Data Manager, Operations or Programme Director in agencies, Compliance operations leadership	Chanel, Deliveroo
Freelance / expert networks	Privacy subject-matter expert, AI training data expert, privacy consultant (day rate)	Ethos (\$90/hour, remote), Context Recruitment (£550/day, inside IR35)

Search parameters (defaults)

Parameter	Default	Notes
Location	London commutable (Greater London plus a 40-mile radius) and UK remote	EU-remote roles are surfaced but flagged "Europe remote" and scored down slightly; UK-wide on-site roles outside the radius are not surfaced
Working pattern	On site, hybrid, remote all included	Hybrid share shown on the card when the posting states it
Contract types	Permanent, fixed-term, contract (day rate, inside or outside IR35), part-time, freelance	All included; the type is a filter in the dashboard. He has applied to a 0.6 FTE role and a 3-month contract
Seniority	Manager to Head of	Director titles included; Analyst, Officer (junior), Executive and Coordinator titles excluded

Parameter	Default	Notes
Salary	No floor	Show the stated salary or day rate on the card. Postings seen range from £49k to £80k permanent and £550/day contract, so a hard floor would remove roles he has chosen to apply for
Posting age	New in the last 24 hours on each daily run; 7 days on the first run	Reposts are deduplicated by company plus normalised title
Languages	English only	

Source documents

The builder receives four files from Nigel. They are the only facts the system may use about him; the letter generator must never invent experience, employers, dates or qualifications beyond them.

File	What it is	How the system uses it
Nigel_Down_CV_Data_Privacy_Professional_October_2026.docx	Two-page CV: profile paragraph, skills list, roles 2002 to 2026 with accounts	Converted to plain text at setup and stored as the candidate profile. Fed to both the fit scorer and the letter writer. Nigel can replace it from the settings page; every letter records which CV version it used

File	What it is	How the system uses it
Job_specification_data.xlsx	His manual tracker: 25 rows, one per application, with columns Date, Company, Prompt Notes, Job Title, Job Specification, Prompt, LinkedIn Profile, Personal Statement, CV, Cover Letter	Three uses. (1) The LinkedIn Profile and Personal Statement columns are identical on every row: extract once and store with the CV as profile context. (2) The Prompt Notes column is the style guide source; see the Cover letter section. (3) The 25 company and title pairs seed the "already applied" list so the first run does not resurface them; the 24 generated letters are kept as few-shot examples of tone

File	What it is	How the system uses it
Sample_Cover_Letter.docx	The PureGym Group Data Compliance Manager letter, 6 October 2026, in his letterhead format	The .docx export template: header block (name, title, mobile, email), date, "Ref: Company, Role", salutation, four to five paragraphs, "Kind regards", name, footer line
CIPPE_CIPM_AIGP_Certificates.pptx	Three slides, each a scanned certificate image	Not used by the pipeline. Store in the app's private file bucket so Nigel can attach them from the dashboard when a form asks for proof

Setup is a one-off import: the builder writes a script that parses the .docx and .xlsx files into the profile and seed tables, and the dashboard settings page lets Nigel upload a replacement CV later.

Job sources

Use three sources: LinkedIn as the primary, Reed.co.uk as the UK-native second, and Indeed (reached through a Google Jobs aggregator) as the wide net. Neither LinkedIn nor Indeed offers a public search API, so both come through third-party providers; each source sits behind a common **JobSource** interface so a provider can be swapped without touching the pipeline.

Source	How to reach it	Why	Cost (Oct 2026)	Risks and mitigations
LinkedIn (primary)	An Apify LinkedIn Jobs actor run from the scheduler, e.g. bebity/linkedin-jobs-scraper . No login or cookies; reads public listings. Inputs: keyword list, locations, publishedAt past 24 hours, work type (on site/remote/hybrid), employment type, experience level. Returns 42 fields including full description, salary, apply URL, applicant count, company profile	Where Nigel finds most roles today; richest structured data; recruiter and company fields feed the letter	From about \$1 per 1,000 results on pay-per-event actors; Apify gives \$5 free credit a month. Expect 100 to 300 results a day across the keyword set, so under \$10 a month	Actors break when LinkedIn changes markup. Pin the actor version, wrap it in the interface, alert when a run returns zero results, and keep a second actor configured as fallback. Do not store recruiter personal data beyond name and profile URL, and purge it with the job after 90 days
Reed.co.uk (second)	Official Jobseeker API, https://www.reed.co.uk/api/1.0/search , free self-serve key, HTTP basic auth (key as username). Params: keywords, locationName, distanceFromLocation (miles), permanent / contract / temp / partTime / fullTime, minimumSalary, resultsToTake (max 100). Search returns a truncated description; call /jobs/{jobId} for the full text of any job that passes the prefilter	Largest UK-native board for compliance and governance roles; the recruitment agencies in Nigel's tracker (Forsyth Barnes, Context Recruitment, Oliver James) post here; an official, stable API	Free. Rate limit undocumented: stay under one request a second and cache job detail	UK only and keyword-only (no remote flag): run the full keyword set and let the fit scorer handle relevance. Reed's terms require attribution and a link back to the listing: show "via Reed.co.uk" on the card
Indeed, plus Glassdoor and company career sites (wide net)	JSearch on RapidAPI , which wraps Google Jobs results. Params: query ("data protection manager in London, UK"), date_posted=today, remote_jobs_only, employment_types. Returns normalised jobs with job_publisher (Indeed, Glassdoor, LinkedIn, employer site) and apply links	Indeed is the highest-volume UK board and has had no public search API since 2024; Google Jobs is the cleanest way in and also catches roles posted only on employer career pages (Regent's University, St John & Elizabeth Hospital, Help for Heroes were all direct postings)	Free tier 200 requests a month; Pro \$25 a month for 10,000. With about 20 queries a day the Pro tier is needed, so budget \$25 a month	Results overlap LinkedIn heavily: dedupe by normalised company plus title plus location before scoring. Google surfaces only what it indexes, so this is a supplement, not the primary

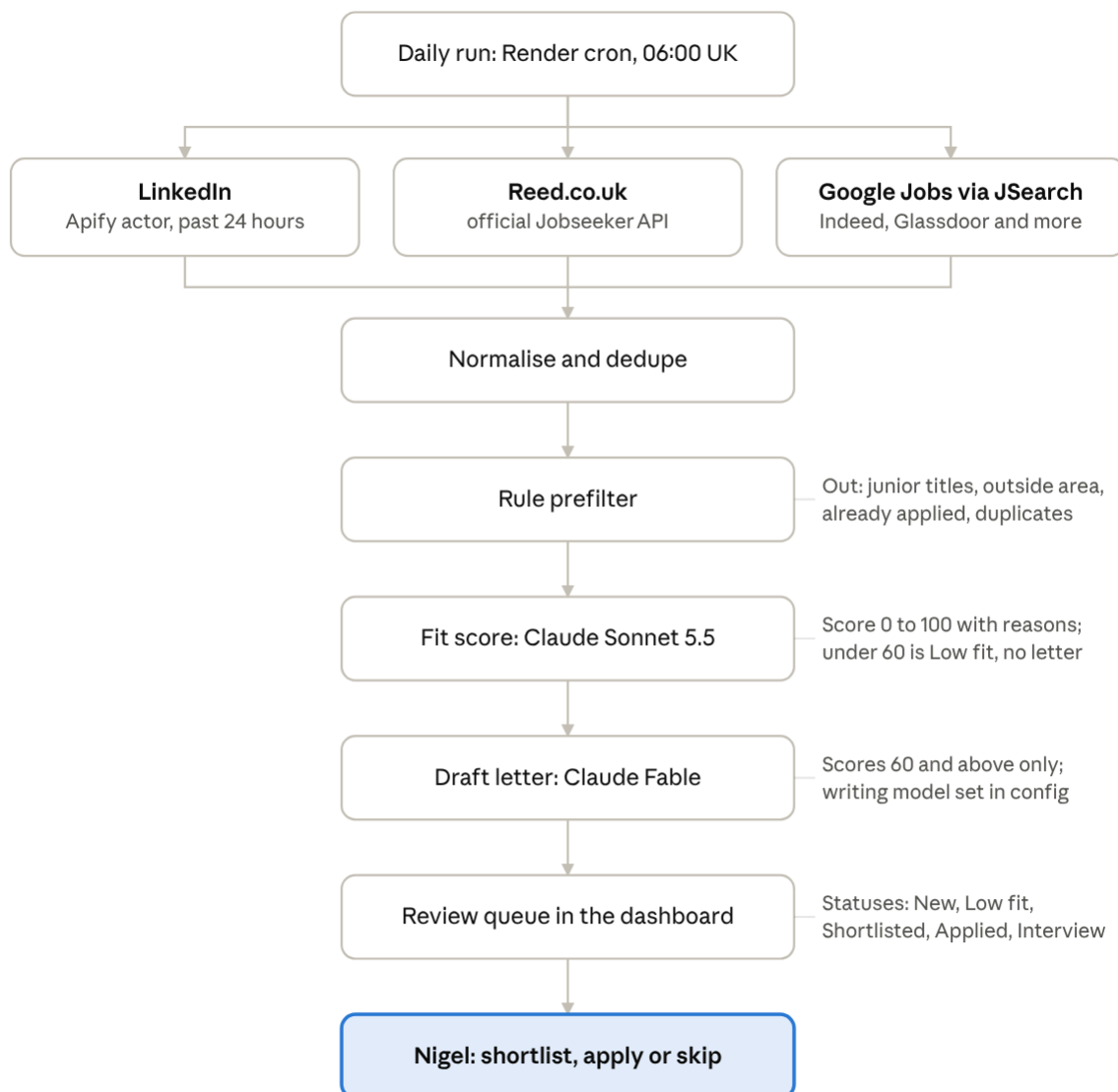
Optional additions for a later phase, in order of value: the [IAPP Career Central](#) job board (low volume, almost every post relevant; no API, so a small HTML fetch with a change alert), and [Adzuna](#) (free key, 250 calls a day, good salary data, aggregates Totaljobs and CV-Library style boards).

Rejected options: logging in to LinkedIn or Indeed with Nigel's own account and automating the browser (account restriction risk and a breach of their user agreements); Proxycurl-style LinkedIn APIs (unavailable since LinkedIn's legal action); scraping Indeed directly (bot detection, constant maintenance).

Search is run as a keyword set, not a single query. The default set is the Core and Adjacent titles from the Target roles table, each combined with the two location settings ("London" with a 40-mile radius; "United Kingdom" with remote only). That is roughly 20 LinkedIn keyword-location pairs in one actor run, 20 Reed calls, and 20 JSearch calls per day.

Pipeline

Nothing reaches an employer without Nigel: find, filter, score, draft, review



daily pipeline · 3 sources, 8 steps, one human gate

One scheduled run a day pulls from three sources, collapses duplicates, drops obvious misses with rules, scores the rest with a cheaper model, drafts letters with the writing model for the good ones, and puts everything in a queue for Nigel.

1. **Schedule.** A Render cron job runs at 06:00 Europe/London daily. The dashboard also has a "Run now" button for Michael. Every run writes a row to `runs`: start, end, per-source counts (fetched, new, filtered, scored, letters), errors, and estimated API cost.
2. **Fetch.** Each source implements `JobSource.search(params): RawJob[]` and is called with the keyword set and both location settings, limited to the past 24 hours (7 days on the first run). Raw payloads are stored as JSON for 30 days for debugging. A source that errors or returns zero results does not stop the run; it is logged and shown as a warning on the dashboard.
3. **Normalise.** Map every source to one `jobs` schema: title, company, location text, remote flag, hybrid days (when stated), salary min and max with currency and period, contract type, posted date, source, listing URL, apply URL, description as plain text. Dedupe key = lower-cased company plus title with punctuation and seniority noise removed plus city. The same company and title seen within 14 days on another source is the same job: keep the first record and add the second source's URL to it. Check every new job against `applications`, seeded with the 25 roles Nigel has already applied for.
4. **Rule prefilter, no model call.** Keep a job if its title contains any Core or Adjacent phrase (or Stretch, when enabled). Drop titles containing Analyst, Officer (except Data Protection Officer), Executive, Coordinator, Assistant (except Assistant Director), Intern, Graduate, Apprentice. Drop on-site roles outside the location radius unless remote. Dropped jobs keep status `filtered` with the reason, visible under a Filtered tab for 30 days so the rules can be tuned.
5. **Fit score.** Claude Sonnet 5.5 (`claude-sonnet-5-5`) scores each surviving job against the profile and returns JSON: `score` 0 to 100, `summary` (two or three sentences), `matches[]`, `gaps[]`, `blockers[]`, `seniority_fit`, `location_fit`, `salary_note`. Rubric: role and responsibilities match 40, must-have requirements 25, sector and context 15, location and working pattern 10, seniority 10. Any hard blocker (qualified solicitor required, on site outside the area, security clearance Nigel lacks) caps the score at 40. Use prompt caching for the profile block, which is identical across calls.
6. **Draft letter.** For scores of 60 and above (threshold in settings), the writing model drafts the letter per the style guide below and returns JSON: `ref_line`, `salutation`, `body_paragraphs[]`, `sign_off`, `notes_for_nigel` (what it emphasised and what in the spec it could not address from his documents). Each letter stores the model ID, prompt version and CV version it used. The .docx export is rendered from the letterhead template with the `docx` npm package.

7. **Queue.** New jobs appear in the dashboard sorted by score; scores under 60 sit under Low fit with no letter, and Nigel can request one with a click.
8. **Notify.** At 07:00 an email (Resend, free tier) lists the day's count and the top five by score with links. Nigel can turn it off in settings.

Cost controls: a per-run cap of 80 scored jobs and 30 letters, and a monthly spend ceiling set in config; when reached, the run completes scoring but skips letters and the dashboard shows why.

Cover letter style guide

The letter is Nigel's voice, not the model's: business-like, friendly and conversational, never gushing, and built only from facts in his documents. These rules are lifted from the Prompt Notes he wrote on all 25 applications and from the letters he kept.

Voice and content

- Tone: professional and conversational, "not too warm". Confident and practical; no flattery of the employer and no exclamation marks. One sentence that shows he read the spec (the company's situation, programme or values) is enough.
- Length: 250 to 350 words in four or five paragraphs. No bullet points, headings or bold.
- Shape: (1) the role, where he saw it, and the headline fit in one or two sentences (ten-plus years, CIPP/E, CIPM and AIGP); (2) experience mapped to the spec's three or four main requirements, using the spec's own vocabulary; (3) one concrete example; (4) where needed, a direct answer to a gap; (5) a short close and thanks.
- Mantle (the dashboard) appears at most once, as supporting evidence, never as the centrepiece. His exact note: "Include work on the Dashboard but not too much emphasis."
- When a spec asks for a qualified lawyer or legal background, include a sentence in this spirit: although not a qualified solicitor, he offers significant knowledge, experience and management of privacy compliance and has worked closely with legal teams throughout.
- Match emphasis to the role: Purview and online behavioural advertising for adtech and technology employers; master data management, data stewardship and business analysis for data-management roles; the trusted-brand angle for luxury and consumer brands; a slightly warmer tone for charities and healthcare.
- Tense: he left MullenLowe in July 2026. Write "In my most recent role" and never "In my current role". He is available immediately; say so when the spec mentions a start date.

- Facts come only from the CV, LinkedIn summary and personal statement. If the spec asks for something his documents do not support (a specific tool, a sector, a language), do not claim it; either leave it out or note it in `notes_for_nigel`.

Hard rules, checked by code after generation

- No em dashes or en dashes anywhere. Use commas, full stops or colons. (His note on every row: "Please do not use any hyphens", meaning the long dashes; ordinary hyphens in words like "privacy-by-design" are fine.)
- UK English throughout: organisation, programme, visualise, prioritise, licence as a noun. His earlier letters mix "visualizes" and "visualises"; the system should not.
- Certifications written as "CIPP/E, CIPM and AIGP".
- Salutation "Dear Hiring Manager," unless the spec names a person. Sign-off "Kind regards," then "Nigel Down".
- No clichés: passionate, hit the ground running, dynamic, synergy, leverage, delve, I am excited.
- Word count 220 to 380; a letter outside it is regenerated once, then flagged.

Letterhead and .docx layout

From Sample_Cover_Letter.docx: name line "Nigel Down", second line "Data Privacy Professional", third line "m: 07795633377 email: nigel@nigeldown.com"; blank line; the date as "6 October 2026"; "Ref: PureGym, Group Data Compliance Manager" (comma, not a dash); the salutation; the body; "Kind regards,"; "Nigel Down". The dashboard shows the body as plain text for copying into web forms and offers the .docx download.

The AI disclaimer line

Nigel's CV and letters end with: "AI-assisted editing tools were used solely to improve spelling, grammar and readability; all content, experience and achievements are my own." When the system drafts the whole letter that sentence would not be true.

Recommendation: replace it in generated letters with "This letter was drafted with AI assistance and reviewed and edited by me; all experience and achievements are my own", with a per-letter toggle to remove the line entirely. Build the toggle; the wording is Nigel's to confirm (see Open decisions).

Per-job notes

Nigel's tracker has a Prompt Notes column he edits for each role ("Include experience on Purview", "include that it is a trusted brand"). Keep that: every job has a free-text Notes for this letter field, and settings has a Standing notes field applied to every letter. Regenerate reads both.

Generation prompt

The builder implements this as a versioned prompt file; the version number is stored on each letter.

SYSTEM

You write cover letters for Nigel Down, a UK data privacy professional. You write in his voice: business-like, friendly, conversational, not warm or effusive. UK English. Four or five paragraphs, 250 to 350 words, no bullets or headings. Never use an em dash or en dash. Use only facts from the profile below; never invent employers, dates, tools, sectors or qualifications. Mention the Mantle dashboard at most once and only as supporting evidence. He left his last role in July 2026, so write about it in the past tense and say he is available immediately when relevant. If the job requires a qualified solicitor, say plainly that he is not one and explain what he offers instead. Return JSON only, matching the schema.

```
<profile> {CV text} {LinkedIn summary} {personal statement} </profile>
<style_examples> {two of his kept letters, e.g. Colt and Help for Heroes}
</style_examples>
<standing_notes> {settings.standing_notes} </standing_notes>
```

USER

```
<job> company: {company} title: {title} location: {location} contract:
{contract_type} source: {source} spec: {description_plain} </job>
<fit_assessment> {score JSON from the scoring step} </fit_assessment>
<notes_for_this_letter> {job.letter_notes} </notes_for_this_letter>
```

```
Schema: { "ref_line": string, "salutation": string, "body_paragraphs":
string[], "sign_off": string, "notes_for_nigel": string }
```

After generation a second, cheaper model call (Sonnet 5.5) checks the letter against the profile and lists any claim it cannot find support for; anything listed is appended to `notes_for_nigel` and the letter card shows a "check this" marker.

Dashboard

A single-purpose web app, mobile-first, with five screens. Nigel's daily visit is: open Today, work down the list, copy or download a letter, click Applied. No training, no onboarding, nothing to configure on day one.

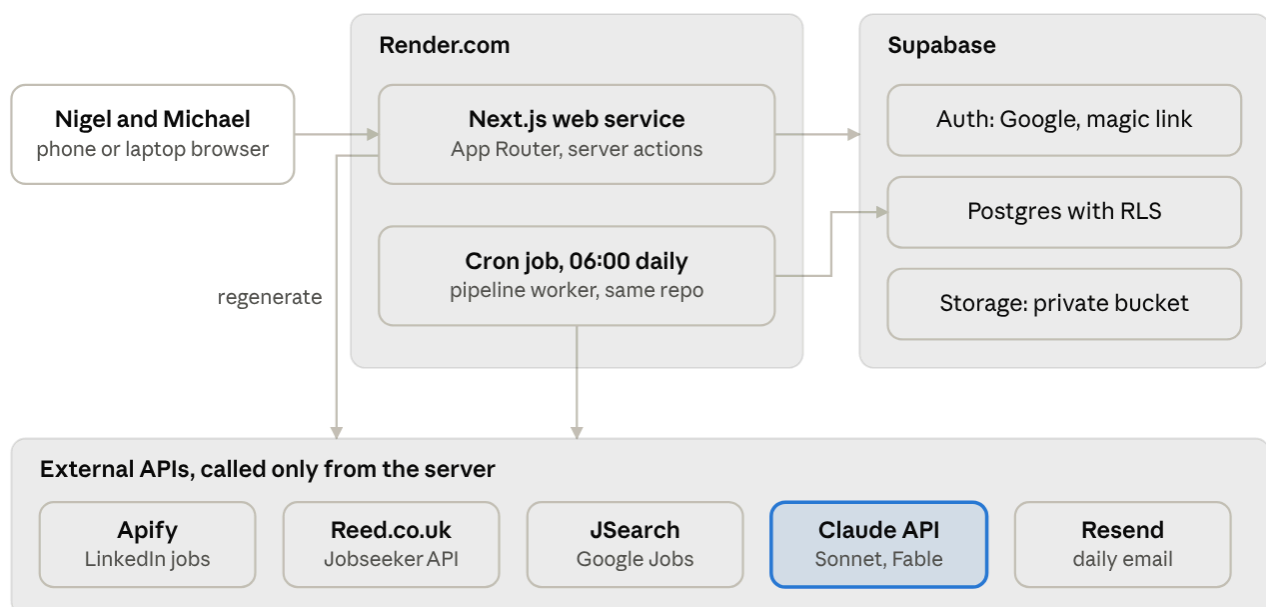
Screen	What it shows	Actions
Login	Sign in with Google (allow-listed) or email magic link. Nothing else on the page	Sign in
Today (home)	New jobs from the latest run, best fit first. Each card: title, company, location and pattern, salary or rate, contract type, source badge with link, posted date, fit score with the two-line summary, blocker warning if any, letter status	Shortlist, Skip, Applied, Open listing, Open letter, Request letter (for Low fit)
Job detail	Full spec on the left (or top on mobile), fit assessment (matches, gaps, blockers), the letter on the right with plain-text copy and .docx download, Notes for this letter field, Regenerate, letter version history	Edit letter inline, Regenerate with notes, change status, add a private note, attach to a calendar date for a deadline
Pipeline	All jobs by status in tabs: New, Shortlisted, Applied, Interview, Offer, Rejected, Skipped, Low fit, Filtered. Search and filter by source, contract type, date	Change status, bulk skip, export the Applied list as CSV (the successor to his spreadsheet)
Settings	Candidate profile (CV upload with version history, LinkedIn summary, personal statement), search keywords by tier with on/off toggles, locations and radius, contract types, score threshold, standing notes, email digest on/off, disclaimer wording and default	Save; changes apply from the next run
Admin (Michael only)	Run history with per-source counts, errors and cost; Run now; API key health; monthly spend against ceiling	Run now, re-run a single source, retry failed letters

States and rules:

- **Statuses:** New, Low fit, Shortlisted, Applied, Interview, Offer, Rejected, Skipped, Filtered, Expired. Applied records the date; Expired is set automatically when a listing's closing date passes or the source reports it closed, and such jobs leave Today.
- A letter is marked Draft until Nigel edits or copies it, then Reviewed. Cards show which.
- **Keyboard and touch:** on mobile, swipe right to shortlist and left to skip; on desktop, J/K to move and S/X/A for shortlist, skip, applied.
- Everything Nigel types (edits, notes) saves automatically and is never overwritten by a regenerate; regenerate creates a new version and keeps the old ones.
- Empty states say what happened ("Run completed 06:04, 143 fetched, 12 new, 7 worth a look") rather than showing a blank page.
- The app works on a phone screen first; the two-column job detail collapses to tabs under 768 px.

Technical architecture

One repo, two Render processes, Supabase for auth, data and files



architecture · Next.js on Render, Supabase, five external APIs

One TypeScript monorepo deployed twice on Render: a web service for the dashboard and a cron job that runs the pipeline. Supabase provides Postgres, auth and private file storage. All third-party API keys live on the server; the browser never calls an external API.

Stack

Layer	Choice	Notes
Framework	Next.js (current stable, App Router), TypeScript strict, React Server Components	Server actions for status changes and regenerate; no separate API server
UI	Tailwind CSS and shadcn/ui	Mobile-first; light and dark follow the system
Database	Supabase Postgres with row-level security; migrations in the repo via the Supabase CLI	Drizzle or the Supabase client, builder's choice
Auth	Supabase Auth: Google provider and email magic link, both restricted to an allow-list table	See Security
Files	Supabase Storage, one private bucket: CV versions, certificates, generated .docx	Signed URLs with a 10-minute expiry
Scheduler	Render cron job running <code>pnpm pipeline</code> on the same image, 06:00 Europe/London	Also invocable from the Admin screen through a server action with a lock to prevent overlap
LLM	<code>@anthropic-ai/sdk</code> . Scoring and fact-check: <code>claude-sonnet-5-5</code> . Writing: <code>WRITING_MODEL</code> env var, default <code>claude-fable-5-1</code> ; set to Fable 5.2's ID when available	Structured JSON output with zod validation; prompt caching on the profile block; retries with backoff
Job sources	<code>apify-client</code> for the LinkedIn actor; <code>fetch</code> for Reed and JSearch; each behind <code>JobSource</code>	Pin the Apify actor version in config
Email	Resend	One template, plain HTML
Documents	<code>docx</code> npm package for the letter export; <code>mammoth</code> and <code>xlsx</code> for the one-off import of Nigel's files	
Observability	Render logs plus a <code>runs</code> table; Sentry free tier optional	Zero-result or failed-source alerts by email to Michael

Data model

Table	Purpose and key columns
allowed_users	email, role (candidate, admin). Sign-in succeeds only for rows here
profile	one row: cv_text, cv_version, linkedin_summary, personal_statement, updated_at
profile_files	storage path, kind (cv, certificate), version, uploaded_at
settings	keyword tiers (JSON), locations and radius, contract types, score_threshold (default 60), standing_notes, digest_enabled, disclaimer_mode and text, monthly_spend_ceiling
runs	started_at, finished_at, trigger (cron, manual), per-source counts (JSON), errors (JSON), estimated_cost_usd
raw_jobs	run_id, source, external_id, payload (JSON); purged after 30 days
jobs	dedupe_key (unique), title, company, location, remote, hybrid_days, salary_min, salary_max, salary_period, currency, contract_type, posted_at, closes_at, sources (JSON array of source plus URL), apply_url, description_text, status, status_changed_at, first_seen_at, expires_at
fit_assessments	job_id, model, prompt_version, score, summary, matches, gaps, blockers (JSON), created_at
letters	job_id, version, model, prompt_version, cv_version, ref_line, salutation, body (JSON array), sign_off, notes_for_nigel, unsupported_claims (JSON), word_count, edited_body (nullable), state (draft, reviewed), docx_path, created_at
job_notes	job_id, text (private notes and the Notes for this letter field)
applications	job_id or free-text company and title for the 25 seed rows, applied_at, outcome, outcome_at
status_events	job_id, from, to, at, by

Configuration

Environment variables, all server-side: `SUPABASE_URL` , `SUPABASE_ANON_KEY` , `SUPABASE_SERVICE_ROLE_KEY` (cron only), `ANTHROPIC_API_KEY` , `WRITING_MODEL` , `SCORING_MODEL` , `APIFY_TOKEN` , `APIFY_LINKEDIN_ACTOR` (actor ID plus version), `REED_API_KEY` , `RAPIDAPI_KEY` , `RESEND_API_KEY` , `DIGEST_TO` , `ALERT_TO` , `MONTHLY_SPEND_CEILING_USD` , `RUN_LOCK_TTL_MINUTES` . Everything a user might change (keywords, threshold, notes) lives in `settings` , not in env.

Repository layout

`app/` (routes and server actions), `pipeline/` (`run.ts` , `sources/linkedin.ts` , `sources/reed.ts` , `sources/jsearch.ts` , `normalise.ts` , `prefilter.ts` , `score.ts` , `write.ts` , `check.ts` , `notify.ts`), `prompts/` (versioned `.md` files), `lib/` (Supabase clients, docx renderer, schemas), `scripts/import-nigel.ts` (one-off import of the four files), `supabase/migrations/` , `tests/` (unit tests for normalise, dedupe and prefilter with fixtures from real listings; a prompt regression test that runs the style checks over generated letters).

Security and privacy

The app holds Nigel's CV, contact details and job-search history, and the owner is a data-protection professional who will notice shortcuts. Treat it as a two-person system with production-grade hygiene.

- Access: Supabase Auth with Google sign-in and email magic link. A sign-in only succeeds for emails in `allowed_users` (`nigel@nigeldown.com` and Michael's address); enforce this in a database trigger on `auth.users` as well as in the app, so a new Google account can never create a session. Sessions expire after 30 days; sign-out everywhere from Settings.
- Roles: `candidate` sees everything except Admin; `admin` sees Admin as well. Both live in `allowed_users` , checked server-side on every request.
- Database: row-level security on every table, with policies that allow reads and writes only for authenticated allow-listed users; the cron job uses the service-role key and runs outside the browser. No table is readable with the anon key.
- Secrets: all API keys in Render environment variables, none in the repo, none sent to the client. The Supabase service-role key is set only on the cron job, not the web service.
- Files: the storage bucket is private; downloads go through short-lived signed URLs generated server-side. Uploads are limited to `.docx` and `.pdf` under 5 MB and are parsed server-side, never served back as-is.

- Transport and headers: HTTPS only (Render default), HSTS, a strict Content-Security-Policy, `SameSite=Lax` cookies, CSRF protection on server actions (Next.js default origin check).
- Third parties: the only data sent outside the system is the profile text and job spec sent to the Claude API for scoring and drafting, and search keywords and locations sent to the job sources. Nothing about Nigel is sent to Apify, Reed or JSearch. Anthropic's API does not train on API inputs by default; the builder should confirm the current data-retention terms at build time and note them in the README.
- Retention: raw source payloads purged after 30 days; filtered jobs after 30 days; expired and skipped jobs after 180 days; letters and applications kept until Nigel deletes them. A Delete all my data action in Settings removes everything and the storage files.
- Recruiter personal data in LinkedIn results (name, profile URL) is stored only on the job record and purged with it; it is never exported.
- Logging: no job descriptions or letters in application logs; log IDs and counts only.
- Dependencies: Dependabot or Renovate on, lockfile committed, `pnpm audit` in CI.

Costs and delivery plan

Expect roughly \$45 a month on a lean setup and about \$110 at full volume (60 jobs scored and 15 letters a day); the Claude API is the largest line and scales with how many letters Nigel wants drafted without asking.

Item	Lean (USD/month)	Full (USD/month)	Basis
Render web service	0 (free instance, cold starts)	7 (Starter, always on)	Render pricing, Oct 2026
Render cron job	1	1	Billed per run minute; under 15 minutes a day
Supabase	0	0	Free tier covers two users, a few thousand rows and under 1 GB of files
Apify (LinkedIn)	0 to 5	10	\$5 free credit; pay-per-result actors from about \$1 per 1,000 jobs, 100 to 300 jobs a day

Item	Lean (USD/month)	Full (USD/month)	Basis
Reed.co.uk API	0	0	Free
JSearch	0 (6 broad queries a day fits the 200-request free tier)	25 (Pro, 10,000 requests)	RapidAPI pricing
Claude API: scoring, Sonnet 5.5	10 (30 jobs a day, cached profile)	20 (60 jobs a day)	\$2 in, \$10 out per million tokens; about 4k input and 400 output tokens a job
Claude API: letters, Fable 5.1	20 (7 letters a day)	45 (15 letters a day)	\$10 in, \$50 out per million tokens; about 6k input and 700 output tokens a letter
Claude API: fact check, Sonnet 5.5	3	8	One short call per letter
Resend	0	0	Free tier, 3,000 emails a month
Total	about 45	about 115	Fable 5.2 pricing unknown; re-estimate if it differs from 5.1

Prompt caching on the profile block (identical in every scoring and writing call) cuts the input side by most of its cost; the figures above assume it is on. The monthly spend ceiling in settings defaults to \$120.

Phases

Phase	Scope	Done when
1. Foundations	Repo, Supabase schema and RLS, auth with allow-list, import script for Nigel's four files, Settings page with profile and keyword tiers	Nigel and Michael can sign in; nobody else can; the profile and 25 seed applications are in the database; <code>pnpm test</code> passes

Phase	Scope	Done when
2. Pipeline	Three sources behind JobSource , normalise and dedupe, prefilter, scoring, Render cron, Admin run history	A manual run fetches from all three sources, writes jobs with scores, and a second run the same day adds no duplicates; a source failure does not abort the run
3. Letters and review	Writing step, style checks, fact check, .docx export, Today and Job detail screens, statuses, mobile layout	Nigel takes a real new job from Today to Applied with a downloaded .docx in under five minutes; ten generated letters pass all hard rules and read, to Nigel, as his own
4. Polish	Email digest, Pipeline screen with CSV export, Filtered tab for rule tuning, spend ceiling, alerts, retention jobs, Delete all my data	A week of unattended daily runs with no manual intervention; the digest arrives at 07:00; spend stays under the ceiling
Later	IAPP and Adzuna sources, CV points to emphasise per role, interview tracker with dates and reminders, Stretch tier search	Only if Nigel asks

Acceptance tests the builder must demonstrate

1. Sign-in with a Google account not on the allow-list is rejected with a clear message and creates no session.
2. The same job posted on LinkedIn and Reed appears once, with both source links.
3. A role requiring a qualified solicitor scores 40 or under and shows the blocker on the card.
4. A generated letter contains no em or en dash, is 220 to 380 words, uses UK spelling, mentions Mantle at most once, and never says "current role".
5. A job the seed list shows as already applied for is never shown in Today.
6. The cron run completes in under 15 minutes and writes a runs row with counts and cost.
7. With WRITING_MODEL changed in Render, the next letter records the new model ID.

Open decisions, assumptions and risks

Three decisions are Nigel's or Michael's to make; the builder should proceed on the defaults below and make each one a setting.

- ☐ Writing model: default `claude-fable-5-1`. Anthropic's model list on 6 October 2026 shows no Fable 5.2; if it has shipped by build time, set `WRITING_MODEL` to its ID and re-check the cost table.
- ☐ AI disclaimer on generated letters: default is the reworded line with a per-letter toggle (see the style guide). Nigel confirms the wording or drops it.
- ☐ Letter threshold: default 60. Nigel can raise it to save cost or lower it to see more drafts.

Assumptions made in this brief, each chosen because Michael expressed no preference:

- Role breadth: Core and Adjacent tiers on by default; Stretch off.
- Location: London plus a 40-mile radius, and UK remote; Europe-remote roles shown but flagged and scored down.
- Contract types: all included, filterable.
- Budget: lean-to-full as in the cost table, with a \$120 monthly ceiling.
- Auth: Google sign-in plus magic link, allow-listed.
- JSearch is the route to Indeed; if Michael prefers a direct Indeed scraper (Apify or Bright Data), swap the adapter.

Risks and how the design handles them:

Risk	Likelihood	Mitigation
The LinkedIn Apify actor breaks or is removed	Likely over a year	<code>JobSource</code> interface, pinned version, a configured fallback actor, zero-result alert to Michael
LinkedIn or Indeed objects to third-party data use	Low for a personal, read-only, no-login tool	No automated applications, no account use, no redistribution of listings
Letters drift from Nigel's voice as the model changes	Medium	Versioned prompts, kept examples, hard-rule checks, the fact-check pass, and Nigel's edits stored for future few-shot use
Model costs spike from a noisy keyword set	Medium	Prefilter before any model call, per-run caps, monthly ceiling

Risk	Likelihood	Mitigation
Nigel stops using it because it surfaces too much	Medium	Threshold and tier toggles in Settings; Low fit tab keeps noise out of Today; Filtered tab lets him tune rules by seeing what was dropped
Reed rate limits or changes its API	Low	One request a second, cached job detail, errors isolated per source
Personal data exposure	Low	Allow-listed auth, RLS, private bucket, no client-side keys, retention rules, delete-all

Sources: [Anthropic models overview](#), [Apify LinkedIn Jobs Scraper](#), [Reed Jobseeker API notes](#), [Free job APIs in 2026](#), [Indeed API status 2026](#), [JSearch overview](#). Nigel's CV, tracker spreadsheet, sample letter and certificates as supplied on 6 October 2026.